

AMAZON CLOUD

AWS Fundamentals and Why?

- AWS provides on-demand delivery of IT resources via the Internet on a secure cloud services platform, offering compute power, storage, databases, content delivery, and other functionality to help businesses scale and grow



Deployment Models

- All-in cloud-based
 - Is fully deployed in the cloud, with all components of the application running in the cloud
- Hybrid Model
 - Is a common approach taken by many enterprises that connects infrastructure and applications between cloud-based resources and existing resources, typically on an existing data center.

Global Infrastructure, Security and Compliance

- AWS serves over one million active customers in more than 190 countries, and it continues to expand its global infrastructure steadily to help organizations achieve lower latency and higher throughput for their business needs.
- AWS provides a highly available technology infrastructure platform with multiple locations worldwide
 - Regions and Availability Zones
- Service Organization Controls (SOC) 1/International Standard on Assurance Engagements (ISAE) 3402, SOC 2, and SOC 3
- Federal Information Security Management Act (FISMA), Department of Defense Information Assurance Certification and Accreditation Process (DIACAP), and Federal Risk and Authorization Management Program (FedRAMP)
- Payment Card Industry Data Security Standard (PCI DSS) Level 1
- International Organization for Standardization (ISO) 9001, ISO 27001, and ISO 27018

Enterprise Applications	 Virtual Desktops	 Sharing and Collaboration
Platform Services	Databases Relational NoSQL Caching Analytics Hadoop Real-Time Data Warehouses Data Workflows App Services Queuing Orchestration App Streaming Transcoding Email Search	Deployment and Management Containers DevOps Tools Resources Templates Usage Tracking Monitoring and Logs Mobile Services Identity Syncs Mobile Analytics Notifications
Foundation Services	 Compute (VMs. Auto Scaling and load Balancing)  Storage (Object, Block and Archive)  Security and Access Control  Networking	
Infrastructure	 Regions  Availability Zones  Content Delivery Networks and Points of Presence	

OVERVIEW ON AWS CLOUD SERVICES

Accessing the Platform

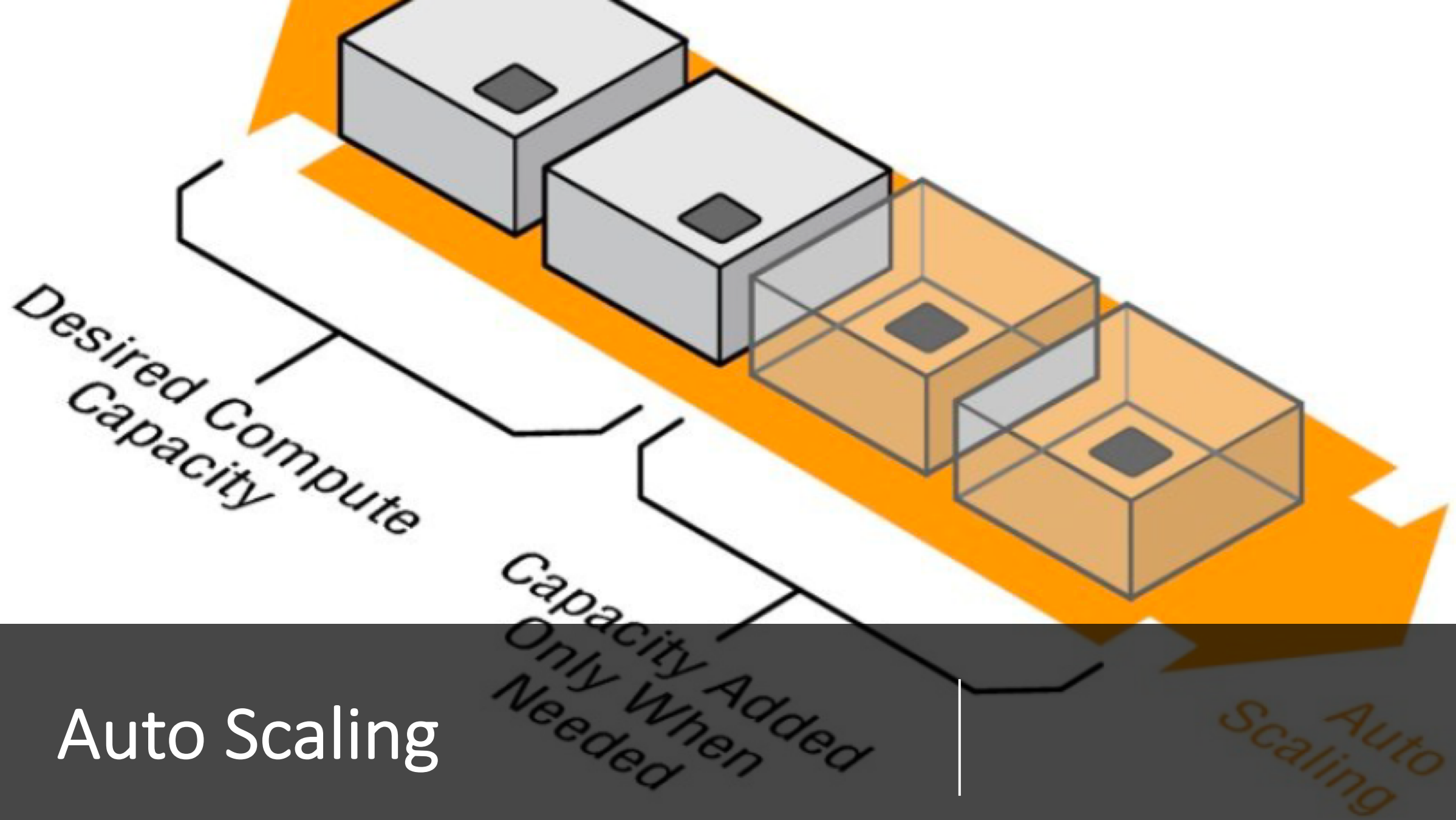
- The AWS Management Console
 - A web application for managing AWS Cloud services.
- The AWS Command Line Interface (CLI)
 - A unified tool used to manage AWS Cloud services
- The AWS Software Development Kits (SDKs)
 - Provide an application programming interface (API) that interacts with the web services that fundamentally make up the AWS platform.

Computing and Networking Services

- AWS provides a variety of compute and networking services to deliver core functionality for businesses to develop and run their workloads.
- The compute and networking services can be leveraged with the storage, database, and application services to provide a complete solution for computing, query processing, and storage across a wide range of applications.

Amazon Elastic Compute Cloud (EC2) & Lambda

- Amazon EC2 presents a true virtual computing environment, allowing organizations to launch compute resources with a variety of operating systems, load them with custom applications, and manage network access permissions while maintaining complete control.
- AWS Lambda runs your back-end code on its own AWS compute fleet of Amazon EC2 instances across multiple Availability Zones in a region, which provides the high availability, security, performance, and scalability of the AWS infrastructure



Elastic Load Balancing and Beanstalk

- Elastic Load Balancing automatically distributes incoming application traffic across multiple Amazon EC2 instances in the cloud. It enables organizations to achieve greater levels of fault tolerance in their applications, seamlessly providing the required amount of load balancing capacity needed to distribute application traffic.
- AWS Elastic Beanstalk is the fastest and simplest way to get a web application up and running on AWS.
- Developers can simply upload their application code, and the service automatically handles all the details, such as resource provisioning, load balancing, Auto Scaling, and monitoring.
- It provides support for a variety of platforms, including PHP, Java, Python, Ruby, Node.js, .NET, and Go.

AWS VPC and Direct Connect

- Amazon Virtual Private Cloud (Amazon VPC) lets organizations provision a logically isolated section of the AWS Cloud where they can launch AWS resources in a virtual network that they define.
- Organizations have complete control over the virtual environment, including selection of the IP address range, creation of subnets, and configuration of route tables and network gateways.
- AWS Direct Connect allows organizations to establish a dedicated network connection from their data center to AWS, which in many cases can reduce network costs, increase bandwidth throughput, and provide a more consistent network experience than Internet-based VPN connections.

OVERVIEW ON STORAGE AND CONTENT DELIVERY

AWS Simple Storage Service (S3)

- Amazon Simple Storage Service (Amazon S3) provides developers and IT teams with highly durable and scalable object storage that handles virtually unlimited amounts of data and large numbers of concurrent users.
- Organizations can store any number of objects of any type, such as HTML pages, source code files, image files, and encrypted data, and access them using HTTP-based protocols.

Amazon Elastic Block Store (EBS)

- Amazon Elastic Block Store (Amazon EBS) provides persistent block-level storage volumes for use with Amazon EC2 instances.
- Each Amazon EBS volume is automatically replicated within its Availability Zone to protect organizations from component failure, offering high availability and durability. By delivering consistent and low-latency performance, Amazon EBS provides the disk storage needed to run a wide variety of workloads.

Amazon CloudWatch

- Amazon CloudWatch is a monitoring service for AWS Cloud resources and the applications running on AWS.
- It allows organizations to collect and track metrics, collect and monitor log files, and set alarms. By leveraging Amazon CloudWatch, organizations can gain system-wide visibility into resource utilization, application performance, and operational health.